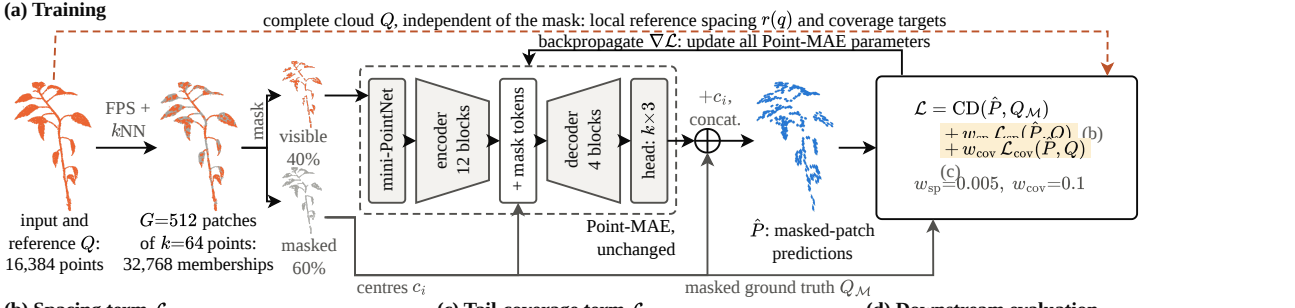

(a) Training
complete cloud Q, independent of the mask: local reference spacing $r(q)$ and coverage targets
backpropagate $\nabla\mathcal{L}$: update all Point-MAE parameters
FPS + kNN
mask
visible 40%
masked 60%
mini-PointNet
encoder 12 blocks
+ mask tokens
decoder 4 blocks
head: $k\times3$
$+c_i$, concat.
$\mathcal{L} = \mathrm{CD}(\hat{P}, Q_{\mathcal{M}})$
$+ w_{\mathrm{sp}}\,\mathcal{L}_{\mathrm{sp}}(\hat{P}, Q)$ (b)
$+ w_{\mathrm{cov}}\,\mathcal{L}_{\mathrm{cov}}(\hat{P}, Q)$ (c)
w_{sp}=0.005, w_{cov}=0.1
input and reference Q: 16,384 points
G=512 patches of k=64 points: 32,768 memberships
Point-MAE, unchanged
$\hat{P}$: masked-patch predictions
centres c_i
masked ground truth $Q_{\mathcal{M}}$
(b) Spacing term $\mathcal{L}_{\mathrm{sp}}$
p
$\phi(\delta)$
crowding
gaps
$1-b$
$1+b$
δ
$\delta = \|p - p'\|/r_p$ to the 4 nearest predictions; grey: no penalty (b=0.25).
(c) Tail-coverage term $\mathcal{L}_{\mathrm{cov}}$
ground truth $q \in Q$
prediction $p \in \hat{P}$
error $e(q; \hat{P})$
worst 10%, $\mathcal{T}_{0.1}$
CD averages all errors; $\mathcal{L}_{\mathrm{cov}}$ averages only the worst 10% ($\mathcal{T}_{0.1}$).CD averages all errors; $\mathcal{L}_{\mathrm{cov}}$ averages only the worst 10% ($\mathcal{T}_{0.1}$).
(d) Downstream evaluation
coverage (recall at τ)
geometric, no segmenter
frozen DGCNN
stem/leaf
gated mIoU
output
Output: $\hat{P}$ + copied visible patches; gated mIoU counts uncovered points as errors.